

Taylor & Francis
Taylor & Francis Group

International Journal of Logistics Research and Applications

A Leading Journal of Supply Chain Management

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/cjol20

Digital product passport for sustainable and circular supply chain management: a structured review of use cases

Abraham Zhang & Stefan Seuring

To cite this article: Abraham Zhang & Stefan Seuring (07 Jul 2024): Digital product passport for sustainable and circular supply chain management: a structured review of use cases, International Journal of Logistics Research and Applications, DOI: 10.1080/13675567.2024.2374256

To link to this article: https://doi.org/10.1080/13675567.2024.2374256

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 07 Jul 2024.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=cjol20

INTERNATIONAL JOURNAL OF LOGISTICS: RESEARCH AND APPLICATIONS
https://doi.org/10.1080/13675567.2024.2374256

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Digital product passport for sustainable and circular supply chain management: a structured review of use cases

Abraham Zhang[a] and Stefan Seuring[b]

[a]Adam Smith Business School, University of Glasgow, Glasgow, UK; [b]Supply Chain Management, University of Kassel, Kassel, Germany

ABSTRACT

Digital product passport (DPP) is a key regulatory element of the latest European Union's Circular Economy Action Plan. By enhancing product traceability and supply chain transparency, its upcoming mandatory implementation aspires to improve resource circularity, cut emissions, and strengthen supply chain governance. However, the extant literature lacks a comprehensive understanding of DPP use cases in the supply chain and how they enable sustainable and circular supply chain management (SCM). To address this knowledge gap, this study first provides an overview of the current state of DPP in policy and practice. Based on a structured review of 82 DPP supply chain use cases in academic literature, we synthesise the functional areas that DPP can add value to improving supply chain sustainability and circularity. Informed by the review findings, we first discuss theoretical and practical implications of DPP-enabled sustainable and circular SCM, and then prioritise research needs in the emerging research domain.

ARTICLE HISTORY
Received 1 May 2023
Accepted 26 June 2024

KEYWORDS
Digital product passport; circular economy; circular supply chain management; sustainable supply chain management; supply chain transparency; use case

1. Introduction

The world faces mounting environmental challenges including increasing pollution, climate change, resource scarcity and depletion, loss of biodiversity, and rapidly growing municipal solid waste, among others. As a response to such challenges, the European Commission (2022a) announced in its European Green Deal to make sustainable products, i.e. products that are environmentally friendly, circular, and energy efficient, the norm in the European Union (EU). Digital product passports (DPPs) will be required for all regulated products on the EU market for enabling a transition to a carbon-neutral and circular economy. A mandatory implementation of DPP will start from industrial and electric vehicle batteries in January 2026 and then expand to more product types (European Commission 2020a; 2022a).

DPP is an electronic tool that registers, processes, and shares product-related information among supply chain entities, government agencies, consumers, and waste management sectors (European Commission 2022b). It captures data from various supply chain and lifecycle stages (van Capelleveen et al. 2023) for improving sustainability performance. For managing environmental sustainability, DPP can collect and provide data about emissions, components, materials, chemical substances, repairability, disassembly, and sustainable disposal instructions for facilitating resource circularity (Adisorn, Tholen, and Götz 2021; Götz et al. 2022). For monitoring social

CONTACT Abraham Zhang abraham.zhang@glasgow.ac.uk Adam Smith Business School, University of Glasgow, 2 Discovery Place, Glasgow G11 6EY, UK

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

sustainability, DPP can provide information related to due diligence in human rights protection at suppliers across the globe (Global Battery Alliance 2023). Essentially, as envisioned by the EU policymakers, DPP helps improve product sustainability and circularity by improving data flows across the supply chain.

As a technological innovation, DPP is still an emerging concept. Understanding its supply chain use cases is crucial for its successful implementation and for evaluating its implications for supply chain research and practice. The use case terminology (Jacobson 2004) is well-established in information systems research, and it is at the heart of 'Industry 4.0' technology adoption (Maghazei, Lewis, and Netland 2022). Essentially, a use case is 'an application of a technology for a specific operational purpose' (Maghazei, Lewis, and Netland 2022). In the early stage of a new technology, use cases are commonly sought for identifying useful application scenarios and for establishing the potential value of the technology (Jacobson 2004).

Against this backdrop, this research investigates specific DPP use cases and their implications in the context of sustainable and circular supply chain management (SCM) (Farooque et al. 2019; Seuring and Müller 2008). The extant literature widely acknowledges the contributions of the circular economy concept to achieving the United Nations sustainable development goals (Vann Yaroson et al. 2024). Circular SCM integrates the circular economy philosophy into the management of the supply chain and its related industrial and ecological systems (Farooque et al. 2019). In comparison with sustainable SCM (Seuring and Müller 2008) which addresses all the three dimensions of the triple bottom line (i.e. environmental, social, and economic), circular SCM focuses on the environmental dimension, more specifically, resource circularity (Zhang et al. 2023a). Hence, we consider circular SCM a subset of the broader sustainable SCM concept.

Academic research on DPP is still at an early stage but substantial growth was observed in the past few years. A few studies report DPP use cases or case studies in various product sectors, for example, electric vehicle batteries (Berger, Schöggl, and Baumgartner 2022) and building materials (Çetin, Gruis, and Straub 2022; Honic, Kovacic, and Rechberger 2019). Nevertheless, we do not have a comprehensive understanding on the current state of DPP research, partly because different names have been used for a variety of DPP variants including cradle-to-cradle passport (Maersk Line 2021), circularity passport (EPEA 2015, as cited in Munaro and Tavares 2021), battery passport (Berger, Schöggl, and Baumgartner 2022), materials passport (Honic, Kovacic and Rechberger, 2019), recycling passport (Spengler and Schröter 2003), among others. There are several review papers covering different aspects related to DPP as summarised in Table 1. Munaro and Tavares (2021) review academic papers addressing material passport implementation challenges and opportunities in the construction sector. Walden, Steinbrecher, and Marinkovic (2021) cover DPP concepts and implementation challenges and opportunities that are found in both academic publications and grey literature. van Capelleveen et al. (2023) review and discuss DPP definitions and related concepts in the academic literature. To the best of our knowledge, however, no review focuses on DPP supply chain use cases which would help inform future research and practice of DPP-enabled sustainable and circular SCM.

Table 1. Comparing this review with other published DPP review papers.

DPP review paper	Scope of review	Review methodology and sample
Munaro and Tavares (2021)	Material passport implementation challenges and opportunities in the construction sector	A systematic review of 15 academic papers
Walden, Steinbrecher, and Marinkovic (2021)	DPP concepts and implementation challenges and opportunities	A narrative review of academic publications and grey literature
van Capelleveen et al. (2023)	DPP definitions and related concepts	A structured review of 200 academic papers related to DPP
This review paper	DPP supply chain use cases in the context of sustainable and circular SCM	A structured review of 75 DPP supply chain use cases found in the academic papers

This review makes a unique contribution to fill in this important knowledge gap by addressing the following research questions:

1.1. How and what potential value can DPP create to enable sustainable and circular SCM?

1.1.1. What are the important research needs in DPP-enabled sustainable and circular SCM?

To answer the first research question, we start with providing an overview of the state-of-the-art of DPP in policy and practice based on an extensive desktop research of grey literature, supplemented by our knowledge of the current state in practice. We then conduct a structured review of DPP supply chain use cases to identify how and what value DPP adds to improving supply chain sustainability and circularity. We search in Scopus, the largest database for academic literature in the business and management discipline. The second research question is addressed by building upon the insights gained from the review results. We first discuss the implications of DPP for supply chain research and practice, and then identify research needs and develop a framework to prioritise them.

The contribution of this review is fourfold. First, it provides a big picture of the current state of DPP in policy and practice which can be useful to readers who are new to the topic. Second, a comprehensive review of DPP supply chain use cases sheds light about how DPP can add value to SCM. Third, based on the review findings, it discusses the implications of DPP for supply chain theory and practice. By enabling restorative and regenerative cycles in the supply chain, DPP is expected to substantially advance supply chain circularity and sustainability. Consequently, we need to elevate the importance of value recovery operations in SCM other than treating them as optional or peripherical as in the past. We need to rethink about the boundaries of the supply chain when it becomes circular, *i.e.* when it no longer ends at the consumption point. Supply chain stakeholders can use DPP in a wide range of functions to improve sustainability and circularity. However, data availability and input can be a major hurdle in the DPP implementation, and it requires close collaboration and incentive alignment among various supply chain stakeholders. Last but not the least, it discusses future research directions about using DPP to advance the sustainability and circularity cause in the supply chain.

The remainder of the paper is organised as follows. Section 2 provides an overview of the current state of DPP in policy and practice. Section 3 reports a structured review of DPP use cases. Based on the review findings, Section 4 discusses the theoretical and practical implications of DPP, and then prioritises research needs in DPP-enabled sustainable and circular SCM. Section 5 concludes the research.

2. DPP in policy and practice

2.1. DPP in policy

The policy push for DPP is evidential in the European Commission's mandate for a transition to a circular and climate-neutral economy. Recently, the European Commission (2019) announced the European Green Deal. It sets long-term goals to make Europe the first climate-neutral continent by 2050 and to decouple economic growth from resource extraction, i.e. to become a circular economy. Under the European Green Deal, a new Circular Economy Action Plan (CEAP) was announced in March 2020 (European Commission 2020a). The new CEAP replaces the old CEAP made in 2015. The new CEAP aims to make the EU economy greener and more competitive, better protect the environment, and empower consumers for more sustainable purchase and consumption. Both the old and the new CEAPs share the same vision of a circular economy, keeping resources in circulation for as long as possible, but they have different implementation strategies. The old CEAP is more concerned about reducing waste sent to landfills and incinerators through closing the

resource loops (European Commission 2015). The new CEAP focuses more on tackling the sources of waste, i.e. product design, production, and consumption activities, aiming to make sustainable products the norm in the EU and empower consumers by providing reliable product sustainability information.

The new CEAP has a distinctive emphasis on the use of digital technologies, which is a manifestation of the EU's green and digital 'Twin Transition' strategy (Muench et al. 2022). It encompasses the following policy elements which all explicitly mandate DPP as part of the solution for transitioning to a circular economy (Protokol 2023).

- **Ecodesign for Sustainable Products Regulation**: The proposed regulation (European Commission 2022b) is an extension of the earlier Ecodesign Directive 2009/125/EC. The latter only covers energy-related products focusing on energy performance. The new regulation proposes to cover resource circularity and other environmental aspects as well. Furthermore, it aims to establish information requirements through DPP for physical goods on the EU market in the future.
- **EU strategy for sustainable and circular textiles**: The strategy sets a 2030 vision that all textile products on the EU market are 'durable, repairable and recyclable, to a great extent made of recycled fibers, free of hazardous substances, produced in respect of social rights and the environment' (European Commission 2022c). DPP will be used to provide better product information so that consumers can verify the green claims of the brands and the regulatory bodies can monitor sustainability compliance.
- **New EU Battery Regulation**. The proposed new battery regulation repeals the Batteries Directive 2006/66/EC which set requirements for the management of waste batteries only. The new regulation expands the scope to cover other lifecycle stages including battery production and use. Article 65 of the new regulation outlines the requirements for a battery passport. By 1 January 2026, each industrial and electric vehicle battery must have a unique and digital battery passport on the EU market (European Commission 2020a).
- **New EU Construction Products Regulation** (CPR). The proposed regulation is a revision of the old CPR Regulation EU 305/2011. In the context of construction products, the revision aims to establish a well-functioning single market and to drive a green and digital transition. DPP will be used to help improve environmental sustainability of construction products and to streamline the administrative works related to data management (European Commission 2022d).

Apparently, DPP is one of the cornerstones of EU's approach to promoting sustainable and circular products (European Commission 2022b). By providing reliable data about product sustainability, DPP is expected to achieve at least four purposes: (1) to enhance transparency about products' environmental impacts at all lifecycle stages; (2) to encourage more sustainable product purchase and consumption; (3) to facilitate product reuse, repair, and recycling; (4) To enable more effective and efficient sustainability compliance management by the public authorities (European Commission 2022b). According to the new CEAP, DPP implementation should be prioritised by considering the circularity potential and environmental impact of different product types. Construction products, industrial and electric vehicle batteries, and textiles are in the first wave of implementation (Protokol 2023). The policy drive is expected to cover other products such as electronics & information and communications technology (ICT), packaging, plastics, and food (European Commission 2020b).

2.2. DPP in practice

Many DPP initiatives have been launched, covering a wide range of products. Jansen et al. (2022) provides a list of 76 DPP projects, all of which are based in Europe. Most of these DPP designs are still at various development stages and have not been implemented or on the market yet. We refer the readers to the list compiled by Jansen et al. (2022) to find the data sources related to these DPP

projects. In this subsection, we provide an up-to-date overview of DPP in practice which is absent in the extant literature.

One of the early and prominent DPP examples is the cradle-to-cradle passport concept adopted by the Maersk Line (2021). Maersk Line publicised the idea of using cradle-to-cradle passport in 2011 for its Triple-E container vessels, the first of which was built in mid-2013. The cradle-to-cradle passport created an online database capturing information on the used materials, including 60,000 tons of steel of seven different grades per ship, and their 3D locations in a ship. It describes how to disassemble and recycle the embedded materials when a ship is to retire after about 30 years in service. The invention provided a solution to overcome a typical problem at ship recycling yards – materials of different grades are mixed up due to the lack of information and they become low-grade, low-price materials (Ellen MacArthur Foundation 2023).

Maersk Line's cradle-to-cradle passport set a new standard for sustainable vessel recycling in the ship building industry. It also popularised the idea of a product passport to a great extent beyond the shipping industry. However, other shipping lines did not follow the best practice. Fortunately, the concept was further explored and more widely adopted by the construction sector. This is not a surprise because the sector has the largest environmental footprint among all industry sectors in the EU, being accountable for about 50% of the total raw material extraction and more than 35% waste generation (Göswein et al. 2022). Therefore, there has been much pressure for the sector to make transition to circular construction. In the building demolition process, the common practice is to crush an old building into rubble which is then sent to landfills. Building material reuse and recycling have been difficult to implement because there is often little data available about the embedded materials in a building several decades after its construction (Tseng et al. 2021).

To overcome the data availability issue, the construction industry, the academia, and several city councils in the EU jointly explored the concept of materials passport. It is worth mentioning the Buildings As Material Banks (BAMB) project funded by the EU's Horizon 2020 research and innovation program (BAMB 2019). The project ran from 2015 to 2019. It developed over 400 materials passports that contained data describing building materials to enable their reuse and recovery. Specifically, the materials passports were designed to provide incentives for suppliers to use circular materials, support reversible building design, make it easier for property developers and renovators to select sustainable and circular building materials, and facilitate efficient and effective recovery of building products, materials, and components (BAMB 2019).

Materials passport usage seems to be accelerating in the European construction industry in recent years. Several commercial players in the EU and in the UK have developed materials passports for their clients in the construction sector. For example, Madaster, based in the Netherlands, provides materials passport solutions. Since its inception in 2017, it has been expanding internationally and now has established operations in Germany, Austria, Belgium, Norway, and Switzerland (Madaster 2023). Construcía (2023), another materials passport provider, has served clients in multiple EU countries. In the UK, Waterman has been pioneering the use of materials passports on London's Edenica building which is still under construction (The Construction Index 2022).

The battery industry has also been active in DPP trial and implementation. The Global Battery Alliance, which is a partnership among the leading battery manufacturers and industry users, governments, academics, non-governmental organisations (NGOs), and international organiaations, has been driving the battery passport adoption since 2019 (Global Battery Alliance 2023). In January 2023, the Global Battery Alliance launched proof-of-concept pilots for the world's first battery passport. The pilots involve a wide range of stakeholders spanning the global battery supply chain from mining to recycling. Some of the main stakeholder groups are automobile manufacturers (Audi, Tesla, and Volkswagen AG), chemical companies (BASF), battery manufacturers (CATL and LG Energy Solution), mining companies (Eurasian Resources Group, Umicore, and Glencore), NGOs (UNICEF, The United Nations Environment Programme (UNEP), and IndustriALL Global Union), IT providers, and governments (the German Ministry for Economic Affairs and Climate Action, and Natural Resources Canada) (Global Battery Alliance 2023).

The Global Battery Alliance has a vision to foster a circular, responsible, and just battery supply chain by 2030 (Global Battery Alliance 2023). In line with this vision, its batter passport contains not only data related to materials to facilitate battery reuse and recycling, but also data on battery carbon footprint and social sustainability due diligence (e.g. in child labour and human rights) in the global supply chain (Global Battery Alliance 2023). Apparently, its battery passport serves a broader sustainability purpose than resource circularity, the key concern of the circular economy concept.

A range of other product sectors have also implemented DPPs. In the textile sector, TrueTwins (2023), a Denmark-based DPP provider, has a large client base of over 150 brands including Cecilie Bahnsen and Mark Cross. Positive Luxury, a certification agency in London, partners with TrueTwins for the implementation of DPP among fashion brands that they certify in meeting the highest sustainability standard (TrueTwins, 2023). iPoint-systems gmbh (2023), a Germany-based sustainability consulting & ICT services firm, offers DPP solutions to industrial production clients. Its DPP deals with both environmental and social sustainability concerns by providing data in responsible sourcing, hazardous substances, recycled content, carbon emissions, product performance and durability, and optimising product 2nd life and beyond. In addition, DPPs have been developed and used for ICT hardware, bikes, and elevators (Holland Circular Hotspot 2021).

In summary, DPP has been piloted in a wide range of product sectors in Europe. Its usage is still in its infancy, but the adoption is accelerating – thanks to the policy push from the European Commission. DPP has many variants, serving different product types and purposes. One of its key purposes is to provide data for achieving resource circularity and environmental sustainability. However, it also serves other purposes related to supply chain social sustainability, compliance management by the regulatory authorities, and product performance.

3. A structured review of DPP supply chain use cases

As mentioned earlier, Munaro and Tavares (2021), Walden, Steinbrecher, and Marinkovic (2021) and van Capelleveen et al. (2023) review research publications related to DPP, but none of them deals with DPP use cases or the implications of DPP for sustainable and circular SCM. For a technological innovation such as DPP, use cases play a crucial role in its adoption for realising its future potential (Maghazei, Lewis, and Netland 2022). This research narrows this important knowledge gap in literature by conducting a review of DPP supply chain use cases that affect sustainability performance. This section first outlines the review methodology, presents the review results, and then summarises DPP supply chain use case areas.

3.1. Review methodology

We conducted a structured literature search in the Scopus database on 18 March 2024. We preferred Scopus to the Web of Science database, another popular choice for literature search, because Scopus has a broader literature coverage (Farooque et al. 2019). The search procedure is outlined in Figure 1. Our Scopus database search targeted publication titles, abstracts, and keywords. To ensure not missing out on any DPP variant, we defined the search keywords based on the very comprehensive and recent review of van Capelleveen et al. (2023). In addition, we focused on finding DPP use cases and case studies related to supply chain sustainability or circularity (Zhang et al. 2021). The search string used was TITLE-ABS-KEY (‘use case’ OR ‘case study’ OR (‘supply chain’ AND (‘circular*’ OR ‘sustainabl*’ OR ‘closed-loop’ OR ‘reverse’ OR ‘remanufactur*’ OR ‘reus*’ OR ‘recycl*’))) AND TITLE-ABS-KEY (‘product passport’ OR ‘material passport’ OR ‘battery passport’ OR ‘building passport’ OR ‘building renovation passport’ OR (‘digital twin’ AND ‘circular economy’) OR (‘digital twin’ AND ‘circularity’) OR (‘digital twin’ AND ‘sustainability’) OR ‘building logbook’ OR ‘C2C passport’ OR ‘Cradle to Cradle passport’ OR ‘Circularity passport’ OR ‘Recycling passport’

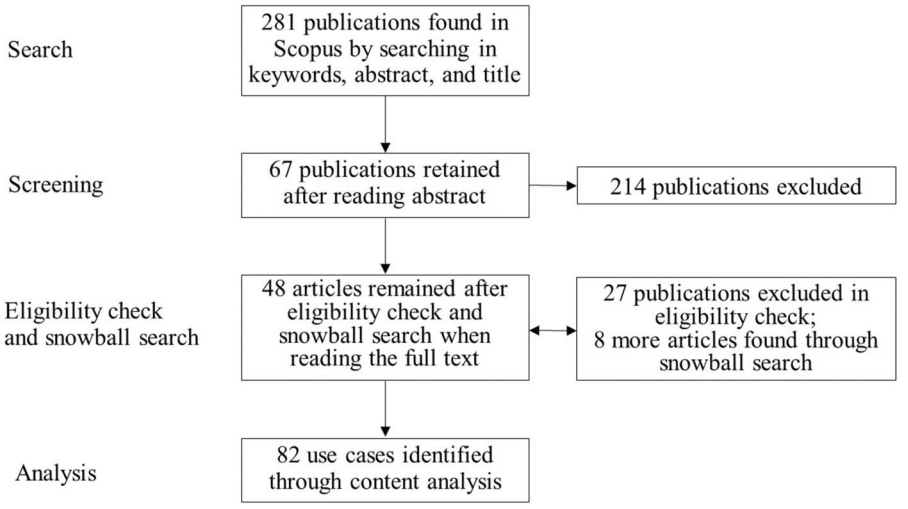


Figure 1. Literature and supply chain use case search procedure.

OR 'nutrient certificate' OR 'Resource passport' OR 'Resource identify tag' OR 'Technical passport for equipment' OR 'Workwear passport') . The search returned 281 publications in English.

The second step was to conduct a screening of the initial sample based on the reading of abstracts. This step excluded 214 publications and retained 67 studies. We enforced three inclusion criteria: (1) A publication must be directly related to DPP (any variant as included in the search keywords is acceptable); (2) The published research must deal with a physical product, not a process or a system; and (3) It must contain a DPP use case (hypothetical or real-life) related to sustainability in one or more supply chain functions or actors. Note that information captured in a digital twin can be related to products as those in a DPP, but it can also be about a process or a system, for example, a production line, farm control, and energy system (van Capelleveen et al. 2023). Our second criteria excluded the studies that were not relevant to a physical product.

The third step was further eligibility check and snowball search. In this step, we read the full text of the publications retained from the second step. We observed that a large portion of these publications on digital twins and virtual reality models focused on mathematical modeling (optimisation and/or simulation). We excluded such studies as they do not deal with DPP use cases or applications. This step excluded 27 publications. While reading the full text, we conducted snowball searches and identified 8 more relevant publications. In total, 48 publications were retained for further analysis.

The last step was to analyze the remaining 48 publications. Some of the publications include more than one use case, but there is a use case being documented in two publications. Some use cases are presented explicitly using the 'use case' terminology, while many others are presented as case studies or opportunities associated with the use of DPP. At last, we arrived at 82 use cases in our final sample. Given that DPP is an emerging topic, we consider such a sample size acceptable. More importantly, unlike a typical review paper that requires a large sample size to survey what was studied in the past, this review is unique by being exploratory in nature. It aims to explore how DPP, as a new technology application, could be used for achieving specific operational purposes in sustainable and circular SCM, informing future research directions.

We performed content analysis (Seuring and Gold 2012) to code all the 82 use cases in the final sample. The analysis process examined six coding categories as outlined in Table 2: stakeholder, useful supply chain application scenario, effect on supply chain sustainability, product type, nature of use case, and country. No theory was used in designing the coding categories as we aimed to

Table 2. Coding categories and coding questions.

Coding category	Coding question
Stakeholder	Which actor is the primary user of DPP in the use case?
Useful supply chain application scenario	Which specific supply chain function/activity does the DPP support and how does it create value?
Effect on supply chain sustainability	Which sustainability dimensions (environmental, social, and/or economic) are affected substantially in the supply chain application?
Product type	Which product type is involved in the DPP use case?
Nature of use case	Does the use case show a proposition (hypothetical/conceptual) that is to be tested, a proof-of-concept (tested in a pilot), or a real-world application (Jansen et al. 2022)?
Country	What is the geographical context of the use case (no matter it is tested/implemented or not)?

obtain descriptive results as a starting point for the later analysis (Potter and Levine-Donnerstein 1999). To ensure the validity of the coding results, we developed a specific coding question as given in Table 2 for each coding category. When evaluating a use case, the rule of coding was to ask the respective coding questions. The answers obtained from text analysis provided inputs into Table A1 (in Appendix). The answers to the coding questions were found for most, but not all, of the use cases. If a use case description does not provide the relevant information, we marked 'unspecified' for the respective category in Table A1.

Given that the coding task dealt with manifest content, which can be observed in the nature of the coding questions, objectivity was relatively easy to obtain by a single coder (Potter and Levine-Donnerstein 1999). While such a process has its shortcomings, the categories are well explained (see Table 2), so that the findings are presented in a transparent manner that could easily be replicated. The single coder also ensures consistency in the analysis of the use cases.

3.2. Review results

There is an obvious upward trajectory in the number of papers published each year on the research topic. As shown in Figure 2, the first publication on the use of DPP in the supply chain was in 2003, ten years before the second publication. After that, research interest on the topic started to grow and accelerated since 2019.

The distribution of publication outlets is presented in Table A1 (in Appendix). The Journal of Cleaner Production has published most (6) papers, followed by the Journal of Building Engineering (4), Procedia CIRP (3), Sustainability (Switzerland) (3), and IOP Conference Series: Earth and

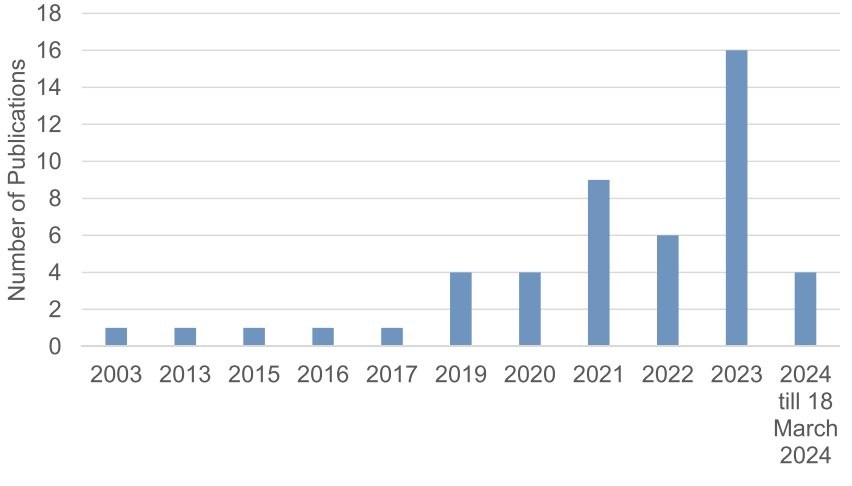


Figure 2. Number of research papers by year.

Environmental Science (2). All the other 30 outlets have published only one paper each. The very diverse range of publication outlets cover academic disciplines spanning across sustainability, building engineering and management, supply chain and operations management, information technology and engineering management, and materials science. This shows that DPP-enabled sustainable and circular SCM requires expertise from multiple academic disciplines.

Table A2 (in Appendix) summarises the supply chain use case review results. A variety of DPP variants, as reflected in the many names for DPPs, appear in the first column. This is partly due to the diverse range of data requirements depending on product characteristics (Berger et al. 2023). For example, 'materials passport' is the most frequently appeared DPP variant and it is widely used for buildings. Construction related products, materials, and wastes have attracted most research attention (42 out of 82 use cases). Other product types include industrial and electric vehicle battery (11), electrical and electronic equipment (4), aerospace product (3), textile and clothing (2), consumer goods (2), ship (2), gas engine (1), cylinder head (1), event equipment (1), tire (1), critical raw materials (1), banana peel (1), post-consumer plastic waste (1), and hydrogen (1). Eight use cases are generic propositions and they do not specify any product type.

There are only 11 real-world DPP applications in the sample and all of them were developed in Europe. Agfa-Gevaert AG, an electrical and electronic equipment manufacturer based in Germany, implemented recycling passports in partnership with its suppliers and recyclers (Spengler and Schröter 2003). Its recycling passports provided product data to recyclers for developing efficient disassembly and recycling strategies. Such use cases, implemented over two decades ago, are the earliest ones that we can find in academic literature. Adisorn, Tholen, and Götz (2021) report Maersk using DPP to enable material reuse and recycling at the end of ship life, which reduces resource demand and cost of building new ships. Adisorn, Tholen, and Götz (2021) also report 'asset administration shell' for systematically recording and retrieving data on manufacturing equipment to aid their management as part of Industry 4.0. Hoosain et al. (2021) report DPP being used to facilitate reuse of sports event equipment for the London 2012 Summer Olympics. The construction sector reports most (seven) real-world DPP use cases (Adisorn, Tholen, and Götz 2021; Brankley et al. 2023; Çetin, Gruis, and Straub 2022; Gómez-Gil et al. 2023; Hoosain et al. 2021). This is not a surprise because the sector led the development of DPP in the past decade.

There are 24 use cases reported in proof-of-concept studies, most of which are related to multiple stakeholders or do not clearly specify a stakeholder. We believe this is mainly because such proof-of-concept pilots were conducted by digital technology experts/researchers, and they had not found an organisation to take ownership of the DPP that's still under development. The remaining 47 use cases are propositional, but they are likely to be valid because they are derived based on the attributes of DPP and commonsense.

European countries dominate the use cases, but there are also proof-of-concept DPPs being developed in China, Israel, Egypt, Kenya, and Brazil. This suggests that non-European countries are joining in the DPP research efforts, and we are likely to see real-world DPP applications being reported outside of Europe in the next few years. Propositional DPP use cases are not tested yet, so many of them are not associated with any country. However, exceptions exist. For example, the BAMB project team, funded by Horizon 2020, developed several use case propositions in the European context (Debacker, Manshoven, and Denis 2016; Luscuere 2017).

3.3. Summary of DPP supply chain use case areas

The useful DPP application scenarios covered in Table A1 are diverse, but some of them overlap to varying degrees. For example, many articles report building materials management for reuse/recycling as a popular application scenario. To gain better clarity in the value of DPP to different stakeholder groups, Table 3 categorises and summarises DPP supply chain use case areas, enabled information flows, and their effect on sustainability.

Table 3. Summary of DPP supply chain use case areas and effect on sustainability.

Stakeholder groups	Supply chain use case areas (number of related publications)	DPP-enabled information flows	Sustainability effect			References
			ENV	SOC	ECO	
Market surveillance authorities	Supply chain sustainability compliance (2)	Information on product environmental and social sustainability from sellers, importers, and producer to market surveillance authorities	+	+		Adisorn, Tholen, and Götz (2021), Berger, Schöggl, and Baumgartner (2022)
Brands and producers	Product design (4)	Information on user needs and environmental performance of design features across internal functions and external supply chain partners	+			Luscuere (2017), Honic, Kovacic, and Rechberger (2019), Adisorn, Tholen, and Götz (2021), Kedir et al. (2023)
	Supply chain development (1)	Information on sustainability and circularity practices and performance between brands/ producers and suppliers	+	+		Berger, Schöggl, and Baumgartner (2022)
	Materials management (8)	Information on products and materials across the supply chain for efficient and effective reuse of materials/components	+	+	+	Hansen, Braungart, and Mulhall (2013), Portillo-Barco and Charnley (2015), 3 XN Adepa (2016 , as cited in Munaro and Tavares 2021), Honic et al. (2019), Munaro et al. (2019), Kovacic, Honic, and Sreckovic (2020), Adisorn, Tholen, and Götz (2021), Psarommatis and May (2024)
	Product authenticity (2)	Information on product origin and certificates from brands and producers to consumers		+	+	Hansen, Braungart, and Mulhall (2013), EPEA (2015; as cited in Munaro and Tavares 2021)
	Remanufacturing (1)	Information on product origin, composition, repair, and dismantling instructions from original manufacturers to remanufacturers	+			Szaller et al. (2023)
Retailers and other sales channels	Product range and assortment planning (1)	Information on product sustainability from manufacturers to sales channels	+			Adisorn, Tholen, and Götz (2021)
	Customer service (1)	Information on product sustainability from manufacturers through sales channels to customers	+			Adisorn, Tholen, and Götz (2021)
Customers/product owners	Sustainable purchase and consumption (3)	Information on product sustainability from brands and producers to consumers	+			Riedelsheimer, Dorfhuber, and Stark (2020), Adisorn, Tholen, and Götz (2021), Berger, Schöggl, and Baumgartner (2022)
	Product modification, maintenance, and repair (9)	Information on product composition, modification, maintenance, and repair instructions from manufacturers to users	+			Portillo-Barco and Charnley (2015), Debacker, Manshoven, and Denis (2016), Riedelsheimer, Dorfhuber, and Stark (2020), Adisorn, Tholen, and Götz (2021)), Sesana et al. (2021), Çetin, Gruis, and Straub (2022), Gonçalves et al. (2022), Gómez-Gil et al. (2023), Çetin et al. (2023)
	Sustainable product disposal (4)	Information on end-of-life product disposal instructions from brands and producers to users	+			Portillo-Barco and Charnley (2015), Riedelsheimer, Dorfhuber, and Stark (2020); Adisorn, Tholen, and Götz (2021), Okumus et al. (2024)
Repair shops	Repair and spare parts management (1)	Information on product composition, modification, maintenance, and repair instructions from manufacturers to users	+		+	Adisorn, Tholen, and Götz (2021)

Secondhand goods markets and secondary materials markets	Buyer trust (1)	Information about how a product has been used from resellers and previous owners/users to customers	+	+	Adisorn, Tholen, and Götz (2021)
	Waste trading (3)	Information on material composition of end-of-life products from users and sellers to buyers and recyclers	+		Hansen, Braungart, and Mulhall (2013), Wu et al. (2023), Lu et al. (2023)
	Material value streams management (1)	Information on the content of secondary materials from sellers and recyclers to users	+	+	EPEA (2015; as cited in Munaro and Tavares 2021)
Recyclers	Building demolition (5)	Information on material composition, toxic or hazardous contents, and product reuse and recycling potential from construction firms to recyclers	+	+	Debacker, Manshoven, and Denis (2016), Çetin, Gruis, and Straub (2022), Maraqa and Spatari (2022), Çetin et al. (2023), Vahidi et al. (2024)
	Waste reuse and recycling (11)	Information on end-of-life product material reuse and recycling from manufacturers to recyclers	+	+	Spengler and Schröter (2003); Hansen, Braungart, and Mulhall (2013), Smeets, Wang, and Drewniok (2019); Adisorn, Tholen, and Götz (2021), Atta, Bakhoum, and Marzouk (2021), Honic et al. (2021), Kovacic and Honic (2021), Berger, Schöggl, and Baumgartner (2022), Plociennik et al. (2022), Akhras et al. (2023), Pourjafarian et al. (2023)
	New business models and partnerships (3)	Information flows between manufacturers and recyclers to facilitate eco-design, end-of-life spare parts management, and value recovery from materials	+	+	Spengler and Schröter (2003), Hansen, Braungart, and Mulhall (2013), Luscuere (2017)
Multiple stakeholders	Product and material traceability (2)	Information on product and material volume, location, and other specifications among multiple stakeholders	+		Gunasekara, Robb, and Zhang (2023); 3 XN Adepa (2016; as cited in Munaro and Tavares 2021)
	Sustainability performance measure and management (4)	Information on sustainability and circularity performance among multiple stakeholders	+		Heisel and Rau-Oberhuber (2020), Boje et al. (2023); Heeß et al. (2024), Soufi, Mesbahi, and Samet (2023)

Note: ENV-Environmental; SOC-Social; ECO-Economical.

As outlined in Table 3, market surveillance authorities such as the European Commission can use DPP to obtain data from sellers, importers, and producer for compliance management of supply chain sustainability. When DPP becomes mandatory, we expect much more efficient and effective compliance monitoring and control as DPP greatly enhances supply chain transparency and product data availability. Such data can cover environmental footprint (e.g. emissions, use of recycled contents, hazard materials) and supply chain social sustainability (e.g. human rights due diligence) (Adisorn, Tholen, and Götz 2021; Berger, Schöggl, and Baumgartner 2022).

Brands and producers can use DPP to facilitate data exchanges across internal functions and external supply chain partners on user needs and environmental performance of design features. Such information flows enable design optimisation, drive design innovation, and strengthen the implementation of circular product design (Kedir et al. 2023; Luscuere 2017). By providing information on sustainability and circularity practices and performance, DPP can assist brands and producers in developing sustainable and circular supply chains jointly with their suppliers (Berger, Schöggl, and Baumgartner 2022). In materials management, DPP can improve traceability of materials including hazardous and critical materials, reduce cost in material/component reuse, and improve supply security by enabling more effective use of secondary raw materials (Adisorn, Tholen, and Götz 2021; Psarommatis and May 2024). By providing information on product origin and certificates from brands and producers to consumers, DPP can help prove product authenticity and guard against counterfeiting, tampering, and adulteration (Hansen, Braungart, and Mulhall 2013). Furthermore, DPP can assist remanufacturing by providing information on product origin, composition, repair, and dismantling instructions from original manufacturers (Szaller et al. 2023).

Retailers and other sales channels can use DPP to acquire data on product sustainability from manufacturers to assist product range and assortment planning (Adisorn, Tholen, and Götz 2021). With sustainability data available in DPP, they can make deliberate choices to increase the sourcing of more sustainable products. They can also use DPP to support customer service by providing sustainability related information to customers (Adisorn, Tholen, and Götz 2021).

Customers can use DPP to easily obtain information on product sustainability from brands and producers to make informed and sustainable purchase decisions (Adisorn, Tholen, and Götz 2021; Berger, Schöggl, and Baumgartner 2022; Riedelsheimer, Dorfhuber, and Stark 2020). Product owners, especially building owners, would benefit from DPP when they carry out product modification, maintenance, and repair (Debacker, Manshoven, and Denis 2016; Gómez-Gil et al. 2023; Portillo-Barco and Charnley 2015). When they decide not to keep the products any longer, they can resell, recycle, or safely dispose of them by following the guidelines provided by brands and producers through DPP (Adisorn, Tholen, and Götz 2021; Portillo-Barco and Charnley 2015; Riedelsheimer, Dorfhuber, and Stark 2020).

Repair shops can use DPP to access information provided by manufacturers on product composition, modification, maintenance, and repair instructions for deciding how to repair a product. Product data at the component level are useful to repair shops for sourcing spare parts (Adisorn, Tholen, and Götz 2021). The growing 'right to repair' movement is likely to benefit from the implementation of DPP.

Secondhand goods marketplaces, for example, eBay, are likely to gain more trust from their customers when DPPs are in place to provide reliable data from resellers and previous owners/users about the transaction and use history of their listed secondhand goods (Adisorn, Tholen, and Götz 2021). Secondary materials markets that trade recycled materials can use DPP to feed information on end-of-life product material composition from users and sellers to buyers and recyclers, improving their control of material value streams (Hansen, Braungart, and Mulhall 2013; Wu et al. 2023). With information available in the DPP on the content of secondary materials from sellers and recyclers, customers can benefit from using secondary materials with known and defined content (EPEA 2015, as cited in Munaro and Tavares 2021).

Recyclers may get a boost from a mandatory implementation of DPP because their operations can become much more efficient when information is available from manufacturers about a product's materials composition, disassembly, reuse, and recycling (Atta, Bakhoum, and Marzouk 2021;

Berger, Schöggl, and Baumgartner 2022). This can be especially true for building demolition which could potentially recover many valuable materials when DPP provides information from construction firms to recyclers on material composition, toxic or hazardous contents, and product reuse and recycling potential (Çetin, Gruis, and Straub 2022; Debacker, Manshoven, and Denis 2016; Maraqa and Spatari 2022). The cost of dismantling will be reduced. Components in good condition can be reused with confidence (Adisorn, Tholen, and Götz 2021). Waste sorting by quality grades can be done more accurately (Gunasekara, Robb, and Zhang 2023; Plociennik et al. 2022). Mixture of materials at different quality levels is likely to be reduced, which will help increase residual value of recycled materials. Consequently, recyclers can increase revenue by selling reusable components and recycled materials at higher quality (Adisorn, Tholen, and Götz 2021). In addition, recyclers may develop new business models by working in partnerships with product designers to improve circular product design and resource recovery. For example, DPP can enable information flows between manufacturers and recyclers to facilitate eco-design, end-of-life spare parts management, and value recovery from materials (Hansen, Braungart, and Mulhall 2013; Luscuere 2017).

4. Discussion

4.1. Theoretical implications

The review findings presented above suggest that the introduction of DPP into sustainable and circular SCM holds significant theoretical implications. First and foremost, we need to redefine the boundaries of SCM and elevate the importance of value recovery operations. We draw Figure 3 to extend the traditional cycle view of the supply chain (Chopra and Meindl 2014, 31–33) to include a restorative cycle for technical materials (e.g. plastic, metal) and a regenerative cycle for biological materials (e.g. food waste) (Farooque et al. 2019). The four cycles of a linear supply chain, namely, procurement cycle, manufacturing cycle, replenishment cycle, and customer order cycle, regulate forward flows of materials and products. The regenerative and restorative cycles, being of a different nature, correspond with reverse flows to close the resource loops. In the traditional SCM concept, a supply chain ends at consumption, but a DPP-enabled supply chain becomes circular and no longer has an end. The DPP concept aligns with the circular economy principles to enable value recovery strategies such as reuse, repair, refurbish, remanufacturing, repurpose, recycling, and composting. Value recovery operations including waste management are pivotal to achieving sustainability and circularity, so they should no longer be considered optional or peripheral in SCM as it has been.

Moreover, the expanded scope of SCM enabled by DPP demands close collaboration of product design and value recovery operations. Some DPP use cases show that most effective and efficient value recovery can only be achieved if product designers intentionally consider facilitating sustainable disposal of end-of-life products (Spengler and Schröter 2003). Given that value recovery operations are often beyond the boundaries of a closed-loop supply chain, theories related to supply chain collaboration need to be augmented to consider the role of complementary supply chains and industry sectors. Therefore, the effects of DPP on supply chain stakeholder relationships can be profound and far-reaching. New theoretical frameworks may be developed to guide the continuous, cradle-to-cradle nature of product and material management in DPP-enabled sustainable and circular SCM.

In addition, the prominent role of DPP in the latest EU's Circular Economy Action Plan underscores the growing importance of interdisciplinary research at the intersection of technology and engineering, sustainability, and SCM. DPP offers a novel approach to enhancing supply chain transparency and traceability, which is crucial for advancing sustainability. From a theoretical perspective, this innovation enables a more data rich, dynamic, and collaborative SCM model to foster a more sustainable and circular economy. Theories related to information systems, technology and engineering management, sustainability, and SCM must keep up with such a development to accommodate and provide guidance to manage the evolving landscape of digital technologies enabled circular supply chain flows (Lagorio et al. 2022; Rehman Khan et al. 2022).

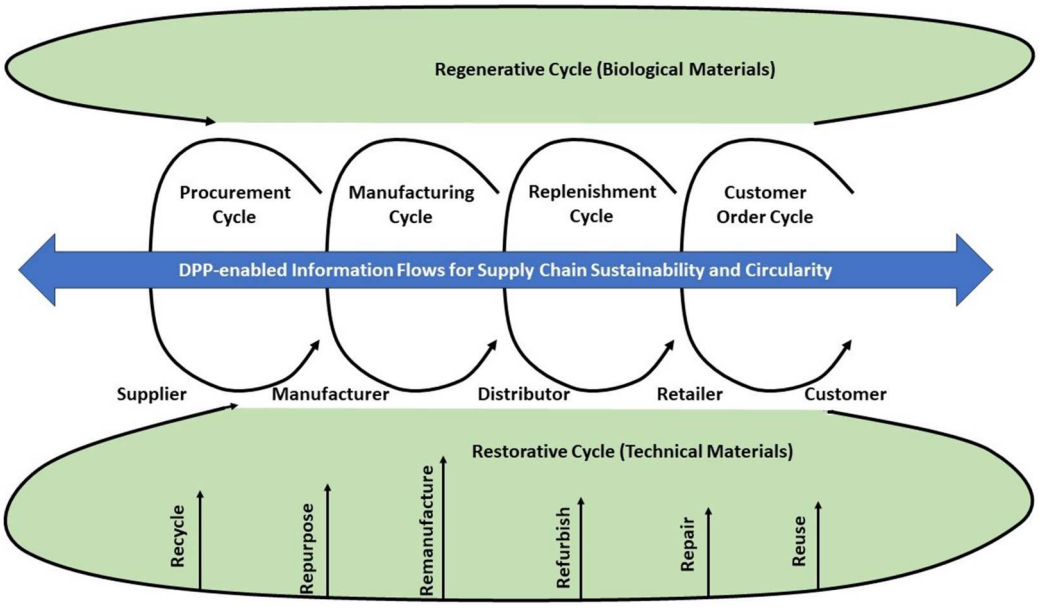


Figure 3. Cycle view of DPP-enabled sustainable and circular SCM. Source: Authors adapted from Chopra and Meindl (2014, 31–33).

4.2. Practical implications

The upcoming mandatory DPP adoption in the EU has implications for a wide range of supply chain stakeholders, which can be inferred from DPP use case areas presented in Table 3. Government authorities in the EU have been studying DPP design and requirements to support their development of DPP related legislation and policies. Given that the circularity pathways of products are dependent on the embedded materials, regional resources availability, and industrial capabilities (Zhang et al. 2021), the policymakers need to develop context-specific guidelines for DPP design and systems architecture. As government authorities drive a transition to a carbon-neutral and circular economy, they need to walk their talk to incorporate sustainability and circularity criteria in public procurement.

Brands and producers will play a major role in the implementation of DPP to enable sustainable and circular SCM. Given that manufacturers are the primary source of product data (Adisorn, Tholen, and Götz 2021), special attention needs to be paid to the study of data requirements and collection for manufacturers by jointly considering DPP information content (Berger et al. 2023). With DPP, it will become much easier for customers to evaluate product sustainability performance. Brands and producers are likely to face heightened competitive pressure to gain legitimacy, win customer trust, and enhance brand reputation.

Suppliers must play an active role in the DPP implementation for improving both environmental and social supply chain sustainability. Apart from components and materials related reporting, emissions management and reporting have been a focus of corporate disclosures. Zhang et al. (2023b)'s recent case studies highlight the need of creating visibility in the emissions data in multi-tier supply chains for accurately reporting and effectively managing scope 3 emissions. Furthermore, multi-tier suppliers' social sustainability has been a topical issue in the past decade, much owing to the recurring problems in the developing countries in the areas of building safety, child labour, minimum wage, and poor working conditions, among others (Venkatesh et al. 2021). However, data requirements and collection at supplier sites can be a major challenge in the implementation of DPP.

Logistics providers are a main contributor to air pollution and greenhouse gas emissions, so emissions management and reporting will be a major concern when they participate in the implementation of DPP along with other supply chain actors. When consumers and corporate customers gain visibility in the employed transport vehicles and their emissions through DPP, they are likely to exert pressure on the logistics providers whose transport activities have a large environmental footprint. Therefore, DPP may trigger an accelerated transition to electric vehicles for green transport and sustainable warehousing (Ibrahim et al. 2022). Furthermore, reverse logistics plays a key part in circular supply chains.

Sales channels involve a wide range of players such as retailers, wholesalers, secondhand goods markets, and secondary raw materials markets, among others, no matter if they operate online or in brick-and-mortar facilities. When DPP is widely implemented, customers can easily compare product information and verify sustainability claims. Sales channels need to respond to the effects of DPP on consumer behaviour to inform their decisions in sourcing, marketing, customer service, and product range and assortment planning.

Customers and consumers will be empowered by DPP to make informed decisions in product purchase and consumption. However, they need to be willing to engage with DPP in their sustainable purchase and end-of-use product disposal decisions. Given that greenwashing has been a concern to many consumers, consumer trust can be a decisive factor determining how much people are willing to use DPP. Moreover, use phase data input can be a major challenge for DPP (Adisorn, Tholen, and Götz 2021).

Value recovery organisations, often classified as in the waste sector, may be transformed in their business models and operations when DPP kicks in. The status quote is a lack of reliable data about the materials composition of products and guidelines for disassembly and recycling. Consequently, limited value is recovered from end-of-use products by the sector and most of the 'wastes' are downcycled or sent to landfills. DPP can be a game changer to enable value recovery organisations to drastically improve efficiency and effectiveness in returns acquisition, sorting, and disposition (Gunasekara, Robb, and Zhang 2023). In addition, DPP may enable value recovery organisations to innovate their business models, for example, to collaborate with manufacturers in circular product design for more effective and efficient recycling operations (Spengler and Schröter 2003).

Looking at the whole supply chain, suppliers, brands, and producers bear more responsibilities in providing data inputs for DPPs, while customers and value recovery operations are more like beneficiaries. Despite different roles and responsibilities, all stakeholders need to work together to make the most out of DPPs to improve supply chain sustainability and circularity for the common good of society (Walden, Steinbrecher, and Marinkovic 2021). Focal firms need to align efforts and rewards to create sufficient incentives for all participating stakeholders in the DPP implementation to overcome circular supply chain barriers (Kazancoglu et al. 2022). Supply chain actors need to collaboratively tackle issues in data availability, cost of data collection, and data authenticity.

Technology providers need to work with industrial users across the supply chain to develop effective technical solutions. Given the large variations in materials composition and user needs, it is likely that different product sectors will have their own DPP design specifications and use requirements. Individual businesses may have the freedom to customise their own DPP designs and systems so long as they meet the mandatory requirements. This presents opportunities as well as challenges to technology providers.

4.3. Future research directions

Based on the theoretical and practical implications presented above, this section discusses research needs related to DPP-enabled sustainable and circular SCM. It is thereby a kind of reflection on the analysis presented in the previous sections. Inspired by the research onion of Saunders, Lewis, and Thornhill (2019), we categorise in Figure 4 a diverse range of research needs by three dimensions: topic, method, and theory.

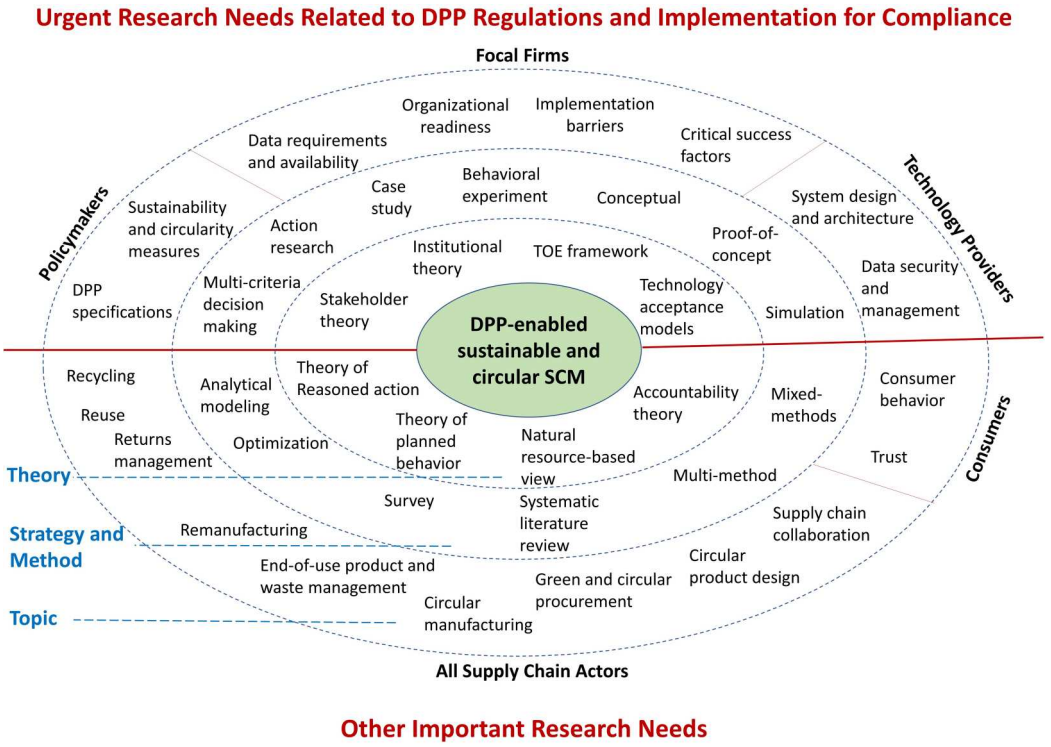


Figure 4. Prioritising research needs in DPP-enabled sustainable and circular SCM.

A wide range of research topics deserve attention, and they are grouped by their respective stakeholders as shown in Figure 4. The top half of the diagram includes the most urgent research needs related to the development of EU DPP regulations and their implementation by businesses to ensure compliance. First, The EU policymakers and the industry need to study and define DPP specifications on the minimum requirements for DPP design and information content, which should vary by product types and their embedded materials. They need to conduct joint research to ensure the upcoming sector-specific regulations are realistic and cost effective to be implemented. Sustainability and circularity measures, which are essential for monitoring compliance, need to be tailored to suit product characteristics (Abdul-Hamid et al. 2022), so applied research is urgently needed to provide clarity to the industry before a mandatory DPP implementation. Second, focal firms need to translate regulatory requirements into their firm and supply chain data requirements and assess data availability (Berger et al. 2023). They need to investigate their organisational readiness, understand DPP implementation barriers, and study critical success factors in order to roll out DPP effectively and efficiently. Third, from a technology provider's perspective, research is needed in system design and architecture to meet the requirements of various user groups. They also need to study different options for data security and management, for example, whether a DPP solution should be built on a centralised database or on a distributed ledger using the blockchain technology.

The bottom half of the diagram encompasses other research needs that are not directly related to the pressing requirements for DPP regulatory compliance. From a user perspective, consumer behaviour and trust are the most prominent concerns, and they are interrelated with each other. However, it is unclear how DPP-enabled supply chain transparency would affect consumer behaviour and trust. Research is also required for understanding how to improve user engagement with DPP not only in sustainable purchase decisions but also in the consumption phase. Finally, it is important to study the implications of DPP for all supply chain actors in improving circularity,

either individually or through supply chain-wide collaboration (Sudusinghe and Seuring 2022). The relevant activities may encompass circular product design (Burke et al., 2023; Wang et al., 2022), green and circular procurement, circular manufacturing, end-of-use product and waste management, remanufacturing, returns management, reuse, and recycling (Farooque et al. 2022; Gunasekara, Robb, and Zhang 2023; Saari et al. 2022). Researchers need to investigate how and under what conditions DPP enables circular SCM through a wide range of disposition options.

The research domain of DPP-enabled sustainable and circular SCM is nascent, so it requires a lot of exploratory studies for understanding the relevant phenomena and underlying theoretical constructs. Proof-of-concept, conceptual, and case studies have dominated the extant literature in DPP, and they will continue to be a good fit before the domain is well-established. Simulation and behavioural experiment methods are useful when empirical data are non-existent or very limited. An action research strategy can be adopted for pilot DPP implementation projects. Multi-criteria decision making is often useful for the studies of barriers and critical success factors and their intertwined relationships (Farooque, Zhang, and Liu 2019). As DPP becomes more widely adopted, survey studies may be conducted for uncovering and testing the causal relationships between theoretical constructs, for example, supply chain transparency, trust, and firm performance. Empirical data can support analytical modeling and optimisation studies to aid sustainable and circular supply chain decisions (Zhang et al. 2021). Mixed-methods and multi-method studies can produce more robust and reliable study results if different data sources are available (Choi, Cheng, and Zhao 2016). As the research domain develops further, systematic literature reviews will be beneficial for surveying the state-of-the-art in literature and for directing further research.

Theories can provide guidance to research endeavours in DPP-enabled sustainable and circular SCM. In the domain's early development stage, institutional theory (DiMaggio and Powell 1983; Shibin et al. 2020) and stakeholder theory (Donaldson and Preston 1995; Siems, Seuring, and Schilling 2023) can be useful for explaining the motivations and drivers of DPP adoption, no matter if it is voluntary or for regulatory compliance. Theories related to technology adoption, for example, technology – organisation – environment (TOE) framework (Baker 2012) and technology acceptance models (Venkatesh and Bala 2008), may offer insights to aid DPP implementation projects. Theory of reasoned action and theory of planned behaviour (Madden, Ellen, and Ajzen 1992) may help with devising intervention strategies, improving consumers' willingness to engage with and willingness to pay for sustainable products. From the viewpoint of firms, natural resource-based view (Hart and Dowell 2011) and accountability theory (Bovens 2007) may offer explanations on the performance outcomes of a DPP implementation. Apart from theory testing and application, there may be opportunities to extend existing theories and to build new theories that can better explain and predict research phenomena related to DPP-enabled sustainable and circular SCM.

5. Conclusion

The world is in dire need of an accelerated transition to a carbon-neutral and circular economy. DPP, as a technological innovation, holds great promise to enable such a transition. DPP is a key regulatory element in the EU's latest Circular Economy Action Plan adopted in 2020. Its mandatory implementation on the EU market will take effect from January 2026 for industrial and electric vehicle batteries, and then expand to more product sectors. DPP can greatly enhance product traceability and supply chain transparency. It is expected to help improve resource circularity, cut emissions, and strengthen supply chain governance. Academic research on DPP has been growing, but there is a lack of comprehensive understanding of DPP supply chain use cases. There is also an urgent need to understand the research needs in DPP-enabled sustainable and circular SCM.

This research makes several academic contributions. First, it provides an overview of the current state of the DPP development in policy and practice, which can be a good reference for both practitioners and researchers. Second, it conducts a structured review of DPP supply chain use cases in academic literature. Our review, being a pioneering one, provides deep insight into how DPP

creates value in a variety of supply chain functions. Third, we discuss theoretical implications based on the review results. The boundaries of SCM need to be extended from the traditional cycle view to include restorative and regenerative cycles for technical and biological materials, respectively. We also need to elevate the importance of value recovery operations, circular product design, supply chain collaboration, and interdisciplinary research. Fourth, we derive practical implications for a wide range of supply chain stakeholders involved in DPP development and implementation. Finally, we discuss research needs in DPP-enabled sustainable and circular SCM. We highlight the most urgent and important research needs in DPP design, circularity measures, implementation barriers, organisational readiness, data security, data requirements and availability, critical success factors, supply chain transparency, trust, and consumer behaviour.

This study has its limitations. It only covers literature in English. Some studies in other languages, for example, German, are omitted from the review work. The scope of our structured review is narrow, focusing on DPP supply chain use cases, but it timely meets the research need at the early stage of the technological innovation. Among the 82 supply chain use cases included in the sample, only 11 of them are real-world applications. This is understandable because a mandatory DPP implementation has not started yet in any product sector in the EU. However, the research domain of DPP-enabled sustainable and circular SCM would benefit from more empirical studies of real-world DPP implementation cases. We specially need to conduct empirical research in non-European contexts because all the reviewed real-world applications were in Europe. As a rapidly developing field, DPP and its applications for sustainable and circular SCM will benefit from an updated review after a few years.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Ministry of Business, Innovation & Employment of New Zealand.

Data availability statement

Data available on request from the authors.

References

Abdul-Hamid, A. Q., M. H. Ali, L. H. Osman, M. L. Tseng, and M. K. Lim. 2022. "Industry 4.0 Quasi-Effect Between Circular Economy and Sustainability: Palm Oil Industry." *International Journal of Production Economics* 253: 108616. https://doi.org/10.1016/j.ijpe.2022.108616.

Adisorn, T., L. Tholen, and T. Götz. 2021. "Towards a Digital Product Passport Fit for Contributing to a Circular Economy." *Energies* 14 (8): 2289. https://doi.org/10.3390/en14082289.

Akhras, M. H., P. J. Freudenthaler, K. Straka, and J. Fischer. 2023. "From Bottle Caps to Frisbee — A Case Study on Mechanical Recycling of Plastic Waste Towards a Circular Economy." *Polymers* 15 (12): 2685. https://doi.org/10.3390/polym15122685.

Atta, I., E. S. Bakhoum, and M. M. Marzouk. 2021. "Digitizing Material Passport for Sustainable Construction Projects Using BIM." *Journal of Building Engineering* 43: 103233. https://doi.org/10.1016/j.jobe.2021.103233.

Baker, J. 2012. "The Technology–Organization–Environment Framework." *Information Systems Theory: Explaining and Predicting Our Digital Society* 28: 231–245. https://doi.org/10.1007/978-1-4419-6108-2_12.

BAMB. 2019. *Enabling a circular building industry*. https://www.bamb2020.eu/.

Berger, K., R. J. Baumgartner, M. Weinzerl, J. Bachler, K. Preston, and J. P. Schöggl. 2023. "Data Requirements and Availabilities for a Digital Battery Passport–A Value Chain Actor Perspective." *Cleaner Production Letters* 4: 100032. https://doi.org/10.1016/j.clpl.2023.100032.

Berger, K., J. P. Schöggl, and R. J. Baumgartner. 2022. "Digital Battery Passports to Enable Circular and Sustainable Value Chains: Conceptualization and Use Cases." *Journal of Cleaner Production* 353: 131492. https://doi.org/10.1016/j.jclepro.2022.131492.

Boje, C., ÁJH Menacho, A. Marvuglia, E. Benetto, S. Kubicki, T. Schaubroeck, and T. N. Gutiérrez. 2023. "A Framework Using BIM and Digital Twins in Facilitating LCSA for Buildings." *Journal of Building Engineering* 76: 107232. https://doi.org/10.1016/j.jobe.2023.107232.

Bovens, M. 2007. "Analysing and Assessing Accountability: A Conceptual Framework." *European Law Journal* 13 (4): 447–468. https://doi.org/10.1111/j.1468-0386.2007.00378.x.

Brankley, L., L. Camci, A. Tugrul, and A. Woolnough. 2023. "Digitalisation of Reinforcing Steel Supply Chain in Construction." International Symposium of the International Federation for Structural Concrete, 1664–1672. Cham: Springer Nature Switzerland.

Burke, H., A. Zhang, and J. X. Wang. 2023. "Integrating Product Design and Supply Chain Management for a Circular Economy." *Production Planning & Control* 34 (11): 1097–1113. https://doi.org/10.1080/09537287.2021.1983063.

Çetin, S., V. Gruis, and A. Straub. 2022. "Digitalization for a Circular Economy in the Building Industry: Multiple-Case Study of Dutch Social Housing Organizations." *Resources, Conservation & Recycling Advances* 15: 200110. https://doi.org/10.1016/j.rcradv.2022.200110.

Çetin, S., D. Raghu, M. Honic, A. Straub, and V. Gruis. 2023. "Data Requirements and Availabilities for Material Passports: A Digitally Enabled Framework for Improving the Circularity of Existing Buildings." *Sustainable Production and Consumption* 40: 422–437. https://doi.org/10.1016/j.spc.2023.07.011.

Choi, T.-M., T. C. E. Cheng, and X. Zhao. 2016. "Multi-Methodological Research in Operations Management." *Production and Operations Management* 25: 379–389. https://doi.org/10.1111/poms.12534.

Chopra, S., and P. Meindl. 2014. *Supply Chain Management – Strategie, Planung und Umsetzung*. 5th ed., 31–33. Hallbergmoos: Pearson.

Construcía. 2023. "Pioneers in Circular Construction." https://www.construcia.com/en/.

The Construction Index. 2022. "Materials Passports Pioneered on London's Edenica Building." https://www.theconstructionindex.co.uk/news/view/materials-passports-pioneered-on-londons-edenica-building.

Debacker, W., S. Manshoven, and F. Denis. 2016. "D1 Synthesis of the State-of-the-Art: Key Barriers and Opportunities for Materials Passports and Reversible Building Design in the Current System." *BAMB Horizon 2020*. http://www.bamb2020.eu/wp-content/uploads/2016/03/D1_Synthesis-report-on-State-of-the-art_20161129_FINAL.pdf.

DiMaggio, P. J., and W. W. Powell. 1983. "The Iron Cage Revisited: Institutional Isomorphism and Collective Rationality in Organisational Fields." *American Sociological Review* 48 (2): 147–160. https://doi.org/10.2307/2095101.

Donaldson, T., and L. E. Preston. 1995. "The Stakeholder Theory of the Corporation: Concepts, Evidence, and Implications." *The Academy of Management Review* 20 (1): 65–91. https://doi.org/10.2307/258887.

Ellen MacArthur Foundation. 2023. "Using Product Passports to Improve the Recovery and Reuse of Shipping Steel: Maersk Line." https://ellenmacarthurfoundation.org/circular-examples/using-product-passports-to-improve-the-recovery-and-reuse-of-shipping-steel.

European Commission. 2015. "Closing the Loop - An EU Action Plan for the Circular Economy." https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52015DC0614.

European Commission. 2019. "The European Green Deal Sets Out How to Make Europe the First Climate-Neutral Continent by 2050, Boosting the Economy, Improving People's Health and Quality of Life, Caring for Nature, and Leaving No One Behind." https://ec.europa.eu/commission/presscorner/detail/en/ip_19_6691.

European Commission. 2020a. "Concerning Batteries and Waste Batteries, Repealing Directive 2006/66/EC and Amending Regulation (EU) No 2019/1020." https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020PC0798.

European Commission. 2020b. "Changing How We Produce and Consume: New Circular Economy Action Plan Shows the Way to a Climate-Neutral, Competitive Economy of Empowered Consumers." https://ec.europa.eu/commission/presscorner/detail/en/ip_20_420.

European Commission. 2022a. "Green Deal: New Proposals to Make Sustainable Products the Norm and Boost Europe's Resource Independence." https://ec.europa.eu/commission/presscorner/detail/en/ip_22_2013.

European Commission. 2022b. "Ecodesign for Sustainable Products." https://commission.europa.eu/energy-climate-change-environment/standards-tools-and-labels/products-labelling-rules-and-requirements/sustainable-products/ecodesign-sustainable-products_en.

European Commission. 2022c. "EU Strategy for Sustainable and Circular Textiles." https://environment.ec.europa.eu/publications/textiles-strategy_en.

European Commission. 2022d. "Laying Down Harmonised Conditions for the Marketing of Construction Products, Amending Regulation (EU) 2019/1020 and Repealing Regulation (EU) 305/2011." https://ec.europa.eu/docsroom/documents/49315.

Farooque, M., A. Zhang, and Y. Liu. 2019. "Barriers to Circular Food Supply Chains in China." *Supply Chain Management: An International Journal* 24 (5): 677–696. https://doi.org/10.1108/SCM-10-2018-0345.

Farooque, M., A. Zhang, Y. Liu, and J. L. Hartley. 2022. "Circular Supply Chain Management: Performance Outcomes and the Role of Eco-Industrial Parks in China." *Transportation Research Part E: Logistics and Transportation Review* 157: 102596. https://doi.org/10.1016/j.tre.2021.102596.

Farooque, M., A. Zhang, M. Thürer, T. Qu, and D. Huisingh. 2019. "Circular Supply Chain Management: A Definition and Structured Literature Review." *Journal of Cleaner Production* 228: 882–900. https://doi.org/10.1016/j.jclepro.2019.04.303.

Gallina, V., B. Gal, Á. Szaller, D. Bachlechner, E. Ilie-Zudor, and W. Sihn. 2023. "Reducing Remanufacturing Uncertainties with the Digital Product Passport." In *Manufacturing Driving Circular Economy*, edited by H. Kohl, G. Seliger, and F. Dietrich, 60–67. GCSM 2022. Lecture Notes in Mechanical Engineering. Cham: Springer.

Global Battery Alliance. 2023. "About GBA." https://www.globalbattery.org/.

Gómez-Gil, M., M. M. Sesana, G. Salvalai, A. Espinosa-Fernández, and B. López-Mesa. 2023. "The Digital Building Logbook as a Gateway Linked to Existing National Data Sources: The Cases of Spain and Italy." *Journal of Building Engineering* 63: 105461. https://doi.org/10.1016/j.jobe.2022.105461.

Gonçalves, J. D. S., R. Mateus, J. Dinis Silvestre, A. Pereira Roders, and L. Bragança. 2022. "Building Passport for the Sustainable Conservation of Built Heritage." *Journal of Cultural Heritage Management and Sustainable Development*. https://doi.org/10.1108/JCHMSD-10-2021-0177.

Göswein, V., S. Carvalho, C. Cerqueira, and A. Lorena. 2022. "Circular Material Passports for Buildings–Providing a Robust Methodology for Promoting Circular Buildings." *IOP Conference Series: Earth and Environmental Science* 1122 (1): 012049. https://doi.org/10.1088/1755-1315/1122/1/012049.

Götz, T., H. Berg, M. Jansen, T. Adisorn, D. Cembrero, S. Markkanen, and T. Chowdhury. 2022. "Digital Product Passport: the Ticket to Achieving a Climate Neutral and Circular European Economy?" University of Cambridge Institute for Sustainability Leadership (CISL) and the Wuppertal Institute.

Gunasekara, L., D. J. Robb, and A. Zhang. 2023. "Used Product Acquisition, Sorting and Disposition for Circular Supply Chains: Literature Review and Research Directions." *International Journal of Production Economics* 260: 108844. https://doi.org/10.1016/j.ijpe.2023.108844.

Hansen, K., M. Braungart, and D. Mulhall. 2013. "Resource Repletion, Role of Buildings." In *Sustainable Built Environments*, edited by V. Loftness and D. Haase, 502–525. New York: Springer.

Hart, S. L., and G. Dowell. 2011. "Invited Editorial: A Natural-Resource-Based View of the Firm: Fifteen Years After." *Journal of Management* 37 (5): 1464–1479. https://doi.org/10.1177/0149206310390219.

Heeß, P., J. Rockstuhl, M. F. Körner, and J. Strüker. 2024. "Enhancing Trust in Global Supply Chains: Conceptualizing Digital Product Passports for a Low-Carbon Hydrogen Market." *Electronic Markets* 34 (1): 1–20. https://doi.org/10.1007/s12525-024-00690-7.

Heisel, F., and S. Rau-Oberhuber. 2020. "Calculation and Evaluation of Circularity Indicators for the Built Environment Using the Case Studies of UMAR and Madaster." *Journal of Cleaner Production* 243: 118482. https://doi.org/10.1016/j.jclepro.2019.118482.

Holland Circular Hotspot. 2021. "Manufacturing: the Future is Circular." https://hollandcircularhotspot.nl/wp-content/uploads/2021/06/Manufacturing-the-future-is-circular-April2021-1.pdf.

Honic, M., I. Kovacic, P. Aschenbrenner, and A. Ragossnig. 2021. "Material Passports for the End-of-Life Stage of Buildings: Challenges and Potentials." *Journal of Cleaner Production* 319: 128702. https://doi.org/10.1016/j.jclepro.2021.128702.

Honic, M., I. Kovacic, and H. Rechberger. 2019. "Improving the Recycling Potential of Buildings Through Material Passports (MP): An Austrian Case Study." *Journal of Cleaner Production* 217: 787–797. https://doi.org/10.1016/j.jclepro.2019.01.212.

Honic, M., I. Kovacic, G. Sibenik, and H. Rechberger. 2019. "Data-and Stakeholder Management Framework for the Implementation of BIM-Based Material Passports." *Journal of Building Engineering* 23: 341–350. https://doi.org/10.1016/j.jobe.2019.01.017.

Hoosain, M. S., B. S. Paul, S. M. Raza, and S. Ramakrishna. 2021. "Material Passports and Circular Economy." In *An Introduction to Circular Economy*, edited by L. Liu and S. Ramakrishna, 131–158. Singapore: Springer.

Ibrahim, A., Y. Fernando, M. L. Tseng, and M. K. Lim. 2022. "Low-Carbon Warehousing Practices and Challenges: Insights from Emerging Country." *International Journal of Logistics Research and Applications*. https://doi.org/10.1080/13675567.2022.2145276.

Incorvaja, D., Y. Celik, I. Petri, and O. Rana. 2022. "Circular Economy and Construction Supply Chains." 2022 IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT), 92–99. IEEE.

iPoint-systems gmbh. 2023. "Digital Product Passport." https://www.ipoint-systems.com/solutions/digital-product-passport/.

Jacobson, I. 2004. "Use Cases–Yesterday, Today, and Tomorrow." *Software & Systems Modeling* 3 (3): 210–220. https://doi.org/10.1007/s10270-004-0060-3.

Jansen, M., B. Gerstenberger, J. Bitter-Krahe, H. Berg, J. Sebestyén, and J. Schneider. 2022. *Current Approaches to the Digital Product Passport for a Circular Economy*. Wuppertal: Wuppertal Institute.

Jæger, B., and S. Myrold. 2023. "Textile Industry Circular Supply Chains and Digital Product Passports. Two Case Studies." IFIP International Conference on Advances in Production Management Systems, 350–363. Cham: Springer Nature Switzerland.

Kazancoglu, I., Y. Kazancoglu, A. Kahraman, E. Yarimoglu, and G. Soni. 2022. "Investigating Barriers to Circular Supply Chain in the Textile Industry from Stakeholders' Perspective." *International Journal of Logistics Research and Applications* 25 (4-5): 521–548. https://doi.org/10.1080/13675567.2020.1846694.

Kedir, F., D. M. Hall, S. Brantvall, J. Lessing, A. Hollberg, and R. K. Soman. 2023. "Circular Information Flows in Industrialized Housing Construction: The Case of a Multi-Family Housing Product Platform in Sweden." *Construction Innovation*. https://doi.org/10.1108/CI-08-2022-0199.

Koppelaar, R. H., S. Pamidi, E. Hajósi, L. Herreras, P. Leroy, H. Y. Jung, A. Concheso, et al. 2023. "A Digital Product Passport for Critical raw Materials Reuse and Recycling." *Sustainability* 15 (2): 1405. https://doi.org/10.3390/su15021405.

Kovacic, I., and M. Honic. 2021. "Scanning and Data Capturing for BIM-Supported Resources Assessment: A Case Study." *Journal of Information Technology in Construction* 26 (32): 624–638. https://doi.org/10.36680/j.itcon.2021.032.

Kovacic, I., M. Honic, and M. Sreckovic. 2020. "Digital Platform for Circular Economy in AEC Industry." *Engineering Project Organization Journal* 9: 1–16. https://doi.org/10.25219/epoj.2020.00107.

Lagorio, A., G. Zenezini, G. Mangano, and R. Pinto. 2022. "A Systematic Literature Review of Innovative Technologies Adopted in Logistics Management." *International Journal of Logistics Research and Applications* 25 (7): 1043–1066. https://doi.org/10.1080/13675567.2020.1850661.

Li, Q., and Y. Wang. 2021. "Blockchain's Role in Supporting Circular Supply Chains in the Built Environment. 2021 IEEE International Conference on Blockchain (Blockchain), 578–583. IEEE.

Liu, Y., M. Farooque, C. H. Lee, Y. Gong, and A. Zhang. 2023. "Antecedents of Circular Manufacturing and its Effect on Environmental and Financial Performance: A Practice-Based View." *International Journal of Production Economics* 260: 108866. https://doi.org/10.1016/j.ijpe.2023.108866.

Lu, W., Z. Peng, C. Webster, and L. Wu. 2023. "Developing a Construction Waste Material 'Passport' for Cross-Jurisdictional Trading." *Journal of Cleaner Production* 414: 137509. https://doi.org/10.1016/j.jclepro.2023.137509.

Luscuere, L. M. 2017. "Materials Passports: Optimising Value Recovery from Materials." *Proceedings of the Institution of Civil Engineers-Waste and Resource Management,* 170 (1): 25–28. Thomas Telford Ltd.

Madaster. 2023. "Increasing the Value of Materials - Circular Construction Starts with Madaster." https://madaster.com/.

Madden, T. J., P. S. Ellen, and I. Ajzen. 1992. "A Comparison of the Theory of Planned Behavior and the Theory of Reasoned Action." *Personality and Social Psychology Bulletin* 18 (1): 3–9. https://doi.org/10.1177/0146167292181001.

Maersk Line. 2021. "Maersk Cradle to Cradle Passport - Total Vessel Recycling." http://www.c2c-centre.com/library-item/maersk-cradle-cradle%C2%AE-passport.

Maghazei, O., M. A. Lewis, and T. H. Netland. 2022. "Emerging Technologies and the Use Case: A Multi-Year Study of Drone Adoption." *Journal of Operations Management* 68 (6-7): 560–591. https://doi.org/10.1002/joom.1196.

Maraqa, M. J., and S. Spatari. 2022. "BIM Material Passport to Support Building Deconstruction and a Circular Economy." In *Construction in the 21st Century 12th International Conference (CITC 12)*, 148.

Muench, S., E. Stoermer, K. Jensen, T. Asikainen, M. Salvi, and F. Scapolo. 2022. "Towards a Green and Digital Future." EUR 31075 EN, Publications Office of the European Union, Luxembourg, ISBN 978-92-76-52452-6. https://doi.org/10.2760/54JRC129319.

Munaro, M. R., A. C. Fischer, N. C. Azevedo, and S. F. Tavares. 2019. "Proposal of a Building Material Passport and its Application Feasibility to the Wood Frame Constructive System in Brazil." *IOP Conference Series: Earth and Environmental Science* 225 (1): 012018. https://doi.org/10.1088/1755-1315/225/1/012018.

Munaro, M. R., and S. F. Tavares. 2021. "Materials Passport's Review: Challenges and Opportunities Toward a Circular Economy Building Sector." *Built Environment Project and Asset Management* 11 (4): 767–782. https://doi.org/10.1108/BEPAM-02-2020-0027.

Okumus, D., S. A. Gunbeyaz, R. E. Kurt, and O. Turan. 2024. "An Approach to Advance Circular Practices in the Maritime Industry Through a Database as a Bridging Solution." *Sustainability* 16 (1): 453. https://doi.org/10.3390/su16010453.

Pehlken, A., and S. Baumann. 2021. "No More Flat Tires: Overcoming Data Defects to Achieve Supply Chain Resilience." 2021 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC), 1–7. IEEE.

Plociennik, C., M. Pourjafarian, A. Nazeri, W. Windholz, S. Knetsch, J. Rickert, … A. Weidenkaff. 2022. "Towards a Digital Lifecycle Passport for the Circular Economy." *Procedia CIRP* 105: 122–127. https://doi.org/10.1016/j.procir.2022.02.021.

Portillo-Barco, C., and F. Charnley. 2015. "Data Requirements and Assessment of Technologies Enabling a Product Passport Within Products Exposed to Harsh Environments: A Case Study of a High Pressure Nozzle Guide Vane." *International Journal of Product Lifecycle Management* 8 (3): 253–282. https://doi.org/10.1504/IJPLM.2015.074145.

Potter, W. J., and D. Levine-Donnerstein. 1999. "Rethinking Validity and Reliability in Content Analysis." *Journal of Applied Communication Research* 27 (3): 258–284. https://doi.org/10.1080/00909889909365539.

Pourjafarian, M., C. Plociennik, M. H. Rimaz, P. Stein, M. Vogelgesang, C. Li, et al. 2023. "A Multi-Stakeholder Digital Product Passport Based on the Asset Administration Shell." 2023 IEEE 28th International Conference on Emerging Technologies and Factory Automation, 1–8. IEEE.

Protokol. 2023. "Digital Product Passport (DPP): A Complete Guide." https://www.protokol.com/insights/digital-product-passport-complete-guide/.

Psarommatis, F., and G. May. 2024. "Digital Product Passport: A Pathway to Circularity and Sustainability in Modern Manufacturing." *Sustainability* 16 (1): 396. https://doi.org/10.3390/su16010396.

Rehman Khan, S. A., Z. Yu, S. Sarwat, D. I. Godil, S. Amin, and S. Shujaat. 2022. "The Role of Block Chain Technology in Circular Economy Practices to Improve Organisational Performance." *International Journal of Logistics Research and Applications* 25 (4–5): 605–622. https://doi.org/10.1080/13675567.2021.1872512.

Riedelsheimer, T., L. Dorfhuber, and R. Stark. 2020. "User Centered Development of a Digital Twin Concept with Focus on Sustainability in the Clothing Industry." *Procedia CIRP* 90: 660–665. https://doi.org/10.1016/j.procir.2020.01.123.

Saari, L., J. Heilala, T. Heikkilä, J. Kääriäinen, A. Pulkkinen, and T. Rantala. 2022. *Digital Product Passport Promotes Sustainable Manufacturing: Whitepaper*. Espoofax: VTT Technical Research Centre of Finland.

Saunders, M., P. Lewis, and A. Thornhill. 2019. *Research Methods for Business Students*. 8th ed. Harlow: Pearson.

Sesana, M. M., G. Salvalai, D. Brutti, C. Mandin, and W. Wei. 2021. "ALDREN: A Methodological Framework to Support Decision-Making and Investments in Deep Energy Renovation of Non-Residential Buildings." *Buildings* 11: 3. https://doi.org/10.3390/buildings11010003.

Seuring, S., and S. Gold. 2012. "Conducting Content-Analysis Based Literature Reviews in Supply Chain Management." *Supply Chain Management: An International Journal* 17 (5): 544–555. https://doi.org/10.1108/13598541211258609.

Seuring, S., and M. Müller. 2008. "From a Literature Review to a Conceptual Framework for Sustainable Supply Chain Management." *Journal of Cleaner Production* 16 (15): 1699–1710. https://doi.org/10.1016/j.jclepro.2008.04.020.

Shibin, K. T., R. Dubey, A. Gunasekaran, B. Hazen, D. Roubaud, S. Gupta, and C. Foropon. 2020. "Examining Sustainable Supply Chain Management of SMEs Using Resource Based View and Institutional Theory." *Annals of Operations Research* 290: 301–326. https://doi.org/10.1007/s10479-017-2706-x.

Siems, E., S. Seuring, and L. Schilling. 2023. "Stakeholder Roles in Sustainable Supply Chain Management: A Literature Review." *Journal of Business Economics* 93 (4): 747–775. https://doi.org/10.1007/s11573-022-01117-5.

Siska, V., A. Al-Akrawi, and M. Zackrisson. 2023. "Building a Sustainable Battery Supply Chain with Digital Battery Passports." In *31st Interdisciplinary Information Management Talks, IDIMT 2023*, 347–354.

Smeets, A., K. Wang, and M. P. Drewniok. 2019. "Can Material Passports Lower Financial Barriers for Structural Steel Re-Use?" IOP Conference Series: Earth and Environmental Science, 012006. Brussels: IOP Publishing.

Soufi, C., T. Mesbahi, and A. Samet. 2023. "Digital Battery Passport as an Enabler of Environmental Impact Assessment in Electric Vehicle Applications." 2023 IEEE Vehicle Power and Propulsion Conference, 1–6. IEEE.

Spengler, T., and M. Schröter. 2003. "Strategic Management of Spare Parts in Closed-Loop Supply Chains—a System Dynamics Approach." *Interfaces* 33 (6): 7–17. https://doi.org/10.1287/inte.33.6.7.25179.

Sudusinghe, J. I., and S. Seuring. 2022. "Supply Chain Collaboration and Sustainability Performance in Circular Economy: A Systematic Literature Review." *International Journal of Production Economics* 245: 108402. https://doi.org/10.1016/j.ijpe.2021.108402.

Szaller, Á, V. Gallina, B. Gal, A. Gaal, and C. Fries. 2023. "Quantitative Benefits of the Digital Product Passport and Data Sharing in Remanufacturing." *Procedia CIRP* 120: 928–933. https://doi.org/10.1016/j.procir.2023.09.102.

TrueTwins. 2023. "Transparency with Scale." https://truetwins.com/customers/.

Tseng, M. L., S. Y. Jeng, C. W. Lin, and M. K. Lim. 2021. "Recycled Construction and Demolition Waste Material: A Cost–Benefit Analysis Under Uncertainty." *Management of Environmental Quality: An International Journal* 32 (3): 665–680. https://doi.org/10.1108/MEQ-08-2020-0175.

Vahidi, A., A. T. Gebremariam, F. Di Maio, K. Meister, T. Koulaeian, and P. Rem. 2024. "RFID-based Material Passport System in a Recycled Concrete Circular Chain." *Journal of Cleaner Production* 442: 140973. https://doi.org/10.1016/j.jclepro.2024.140973.

van Capelleveen, G., D. Vegter, M. Olthaar, and J. van Hillegersberg. 2023. "The Anatomy of a Passport for the Circular Economy: A Conceptual Definition, Vision and Structured Literature Review." *Resources, Conservation & Recycling Advances* 17: 200131.

Vann Yaroson, E., S. Chowdhury, S. K. Mangla, P. Dey, F. T. Chan, and M. Roux. 2024. "A Systematic Literature Review Exploring and Linking Circular Economy and Sustainable Development Goals in the Past Three Decades (1991–2022)." *International Journal of Production Research* 62 (4): 1399–1433. https://doi.org/10.1080/00207543.2023.2270586.

Venkatesh, V., and H. Bala. 2008. "Technology Acceptance Model 3 and a Research Agenda on Interventions." *Decision Sciences* 39 (2): 273–315. https://doi.org/10.1111/j.1540-5915.2008.00192.x.

Venkatesh, V. G., A. Zhang, E. Deakins, and V. Mani. 2021. "Antecedents of Social Sustainability Noncompliance in the Indian Apparel Sector." *International Journal of Production Economics* 234: 108038. https://doi.org/10.1016/j.ijpe.2021.108038.

Walden, J., A. Steinbrecher, and M. Marinkovic. 2021. "Digital Product Passports as Enabler of the Circular Economy." *Chemie Ingenieur Technik* 93 (11): 1717–1727. https://doi.org/10.1002/cite.202100121.

Wandiga, C. 2020. "Methodological Review: Socio-Cultural Analysis Criteria for BIM Modeling and Material Passport Tracking of Agriwaste as a Building Construction raw Material." *MRS Energy & Sustainability* 7: E25. https://doi.org/10.1557/mre.2020.29.

Wang, J. X., H. Burke, and A. Zhang. 2022. "Overcoming Barriers to Circular Product Design." *International Journal of Production Economics* 243: 108346. https://doi.org/10.1016/j.ijpe.2021.108346.

Wilson, S., K. Adu-Duodu, Y. Li, R. Sham, Y. Wang, E. Solaiman, C. Perera, and O. Rana. 2023. "Tracking Material Reuse Across Construction Supply Chains." 2023 IEEE 19th International Conference on e-Science (e-Science), 1–4. IEEE.

Wu, L., W. Lu, Z. Peng, and C. Webster. 2023. "A Blockchain non-Fungible Token-Enabled 'Passport' for Construction Waste Material Cross-Jurisdictional Trading." *Automation in Construction* 149: 104783. https://doi.org/10.1016/j.autcon.2023.104783.

Zhang, A., L. Duong, S. Seuring, and J. L. Hartley. 2023a. "Circular Supply Chain Management: A Bibliometric Analysis-Based Literature Review." *The International Journal of Logistics Management*. https://doi.org/10.1108/IJLM-04-2022-0199.

Zhang, A., H. L. Tay, M. F. Alvi, J. X. Wang, and Y. Gong. 2023b. "Carbon Neutrality Drivers and Implications for Firm Performance and Supply Chain Management." *Business Strategy and the Environment* 32 (4): 1966–1980. https://doi.org/10.1002/bse.3230.

Zhang, A., J. X. Wang, M. Farooque, Y. Wang, and T. M. Choi. 2021. "Multi-Dimensional Circular Supply Chain Management: A Comparative Review of the State-of-the-Art Practices and Research." *Transportation Research Part E: Logistics and Transportation Review* 155: 102509. https://doi.org/10.1016/j.tre.2021.102509.

Appendix

Table A1. Distribution of publication outlets.

Publication outlet	Frequency
Journal of Cleaner Production	6
Journal of Building Engineering	4
Procedia CIRP	3
Sustainability (Switzerland)	3
IOP Conference Series: Earth and Environmental Science	2
Polymers	1
Electronic Markets	1
Construction Innovation	1
Energies	1
Lecture Notes in Mechanical Engineering	1
Engineering Project Organization Journal	1
Resources, Conservation and Recycling Advances	1
IDIMT 2023: New Challenges for ICT and Management – 31st Interdisciplinary Information Management Talks	1
Automation in Construction	1
IFIP Advances in Information and Communication Technology	1
An Introduction to Circular Economy	1
Interfaces	1
MRS Energy and Sustainability	1
International Journal of Product Lifecycle Management	1
Buildings	1
Sustainable Built Environments	1
Built Environment Project and Asset Management	1
Sustainable Production and Consumption	1
BAMB Report	1
Proceedings – 2021 IEEE International Conference on Blockchain	1
Proceedings – 2022 IEEE/ACM 9th International Conference on Big Data Computing, Applications and Technologies	1
Proceedings – 2017 Conference of the Institution of Civil Engineers-Waste and Resource Management	1
Proceedings – 2022 International Conference on Construction in the twenty-first Century	1
Proceedings – 2021 IEEE International Conference on Engineering, Technology and Innovation	1
Proceedings – 2023 IEEE 19th International Conference on e-Science	1
2023 IEEE Vehicle Power and Propulsion Conference, VPPC 2023 – Proceedings	1
Journal of Cultural Heritage Management and Sustainable Development	1

Table A2. DPP use case review results.

DPP variant and reference	Stakeholder	Supply chain activity	Affected sustainability dimension			Product type	Nature of use case	Country
			ENV	SOC	ECO			
Digital battery passport (Soufi, Mesbahi, and Samet 2023)	Multiple stakeholders	Environmental impact assessment	+			Electronic vehicle	Proposition	Europe
Digital product passport (Psarommatis and May 2024)	Manufacturer	Circular supply chain	+	+	+	Smartphone	Proposition	Unspecified
Material passport (Vahidi et al. 2024)	Recycler	Building demolition management focusing on material reuse/recycling	+		+	Recycled concrete	Proof-of-concept	Netherlands
Database (Okumus et al. 2024)	Ship owners	Materials reuse/recycling	+		+	Ship	Proof-of-concept	Unspecified
Digital product passport (Heeß et al. 2024)	Multiple stakeholders	Supply chain decarbonisation	+			Hydrogen	Proposition	Global
Digital product passport (Pourjafarian et al. 2023)	Recycler	Waste sorting	+			Waste electrical and electronic equipment	Proof-of-concept	Germany
Digital product passport (Szaller et al. 2023)	Manufacturer	Remanufacturing	+			Cylinder head	Proof-of-concept	Unspecified
Material passport (Wilson et al. 2023)	Multiple stakeholders	Circular SCM	+			Building	Proof-of-concept	UK
Digital battery passport (Siska, Al-Akrawi, and Zackrisson 2023)	Producer	Circular SCM	+			Electronic vehicle and industrial battery	Proof-of-concept	Europe
Koppelaar et al. (2023)	Repair organisation	Repair	+			Electronic vehicle and industrial battery	Proposition	Unspecified
	Recycler	Recycling	+			Electronic vehicle and industrial battery	Proposition	Unspecified
	Consumer	Sustainable purchase	+	+		Electronic vehicle and industrial battery	Proposition	Unspecified
	Retailer	Product sale	+			Electronic vehicle and industrial battery	Proposition	Unspecified
	Remanufacturer	Remanufacturing	+			Electronic vehicle and industrial battery	Proposition	Unspecified
	Multiple stakeholders	Circular SCM	+			Critical raw materials	Proposition	Europe

(*Continued*)

Table A2. Continued.

DPP variant and reference	Stakeholder	Supply chain activity	Affected sustainability dimension			Product type	Nature of use case	Country
			ENV	SOC	ECO			
Jæger and Myrold (2023)	Brand	Circular SCM	+	+		Textile	Proof-of-concept	Sweden
Gallina et al. (2023)	Producer	Remanufacturing	+		+	Gas engine	Proposition	Europe
Brankley et al. (2023)	Multiple stakeholders	Reinforcing steel SCM	+	+	+	Construction steel	Real-world application	UK
Material passport (Kedir et al. 2023)	Construction firm	Building design and construction	+			Building	Proposition	Sweden
Material passport (Çetin et al. 2023)	Social housing organisations	Building maintenance, renovation, and demolition	+			Building	Proposition	Netherlands
Digital Twin (Boje et al. 2023)	Multiple stakeholders	Life cycle sustainability assessment	+			Building	Proposition	Luxembourg
Digital product passport (Akhras et al. 2023)	Recycler	Mechanical recycling of plastic waste	+			Plastic waste	Proof-of-concept	Austria
Waste material passport (Liu et al. 2023; Wu et al. 2023)	Construction and recycling industries	Construction waste material cross-border trading for material reuse/recycling	+			Construction waste	Proof-of-concept	China
Digital building logbook (Gómez-Gil et al. 2023)	Building owners	Building renovation	+			Building	Real-world application	Spain and Italy
Material passport (Çetin, Gruis, and Straub 2022)	Building owners	Building construction, renovation, maintenance, and demolition for material circularity	+			Building	Real-world application	Netherlands
Digital battery passport (Berger, Schöggl, and Baumgartner 2022)	Original equipment manufacturer	Supply chain development to improve sustainability and circularity	+	+		Electric vehicle battery	Proposition	Unspecified
	Electric vehicle user	Support sustainable purchase decision	+	+				
	Recycler	Support recycling activities	+	+	+			
	Regulatory body	Facilitating sustainable product management efforts	+	+				
Material passport (Maraqa and Spatari 2022)	Unspecified	Building demolition management for material reuse/recycling	+			Construction waste	Proof-of-concept	Israel
Incorvaja et al. (2022)	Multiple stakeholders	Circular SCM	+			Building	Proof-of-concept	UK
Building passport (Gonçalves et al. 2022)	Unspecified	Heritage buildings conservation	+	+	+	Building	Proof-of-concept	Netherlands
Digital lifecycle passport (Plociennik et al. 2022)	Multiple stakeholders	Support electronic waste sorting	+		+	Electronics	Proof-of-concept	Unspecified
Material passport (Atta, Bakhoum, and Marzouk 2021)	Multiple stakeholders	Building materials management for reuse/recycling	+			Building	Proof-of-concept	Egypt
Building renovation passport (Sesana et al. 2021)	Multiple stakeholders	Building renovation to improve energy efficiency	+			Building	Proof-of-concept	Italy
Material passport (Honic et al. 2021)	Unspecified	Building demolition management for material reuse/recycling	+			Construction waste	Proof-of-concept	Austria

(*Continued*)

Table A2. Continued.

DPP variant and reference	Stakeholder	Supply chain activity	Affected sustainability dimension			Product type	Nature of use case	Country
			ENV	SOC	ECO			
Hoosain et al. (2021)	Unspecified	Materials reuse/recycling	+			Building	Proof-of-concept	Netherlands
	Unspecified	Equipment reuse	+	+		Sports event equipment	Real-world application	UK
	Unspecified	Materials reuse/recycling	+			Building	Real-world application	Germany
	Unspecified	Materials reuse/recycling	+			Building	Real-world application	Netherlands
	Unspecified	Material reuse and emissions reduction	+			Building	Real-world application	Netherlands
Material passport (Kovacic and Honic 2021)	Unspecified	Building materials management for reuse/recycling	+			Building	Proof-of-concept	Austria
Li and Wang (2021)	Multiple stakeholders	Circular SCM	+			Building	Proposition	UK
Pehlken and Baumann (2021)	Producer	Recycling	+		+	Tire	Proposition	Europe
Digital product passport (Adisorn, Tholen, and Götz 2021)	Construction and deconstruction firms	Enable reversible building design and facilitate recycling and reuse of construction materials	+			Building	Real-world application	Europe
	Maersk Line	Enable material reuse and recycling at the end of vessel life to reduce resource demand and cost of building new ships	+		+	Ship	Real-world application	Denmark
	Suppliers, manufacturers, industrial users	Manufacturing equipment management aided by 'asset administration shell'	+			Unspecified	Real-world application	Unspecified
	Suppliers, integrators, machine manufacturers and other industrial users	Improve product performance and predictive maintenance	+			Unspecified	Proposition	Unspecified
	Market surveillance authorities	Monitor compliance of manufacturers	+			Unspecified	Proposition	Unspecified
	Retailers	Make product range more sustainable and provide sustainability information at point of sale	+			Consumer goods	Proposition	Unspecified
	Second-hand stores	Increase the trust of buyers in second-hand and remanufactured products	+		+	Consumer goods	Proposition	Unspecified
	Product users	Make more conscious decisions to purchase sustainable products	+			Unspecified	Proposition	Unspecified
	Repair shops	Provide information required for repair and spare parts	+		+	Unspecified	Proposition	Unspecified
	Waste management organisations	Reduce dismantling cost and increase revenue by selling recycled materials at higher qualities	+		+	Unspecified	Proposition	Unspecified
Material passport (Heisel and Rau-Oberhuber 2020)	Unspecified	Support the calculation and evaluation of circularity indicators	+			Building	Proof-of-concept	Switzerland
	Unspecified	Building materials management for reuse/recycling	+			Building		Europe

(*Continued*)

Table A2. Continued.

DPP variant and reference	Stakeholder	Supply chain activity	Affected sustainability dimension			Product type	Nature of use case	Country
			ENV	SOC	ECO			
Digital platform for circular economy (Kovacic, Honic, and Sreckovic 2020)							Proof-of-concept	
Wandiga (2020)	Unspecified	Agriwaste to construction raw material	+			Banana peel	Proposition	Kenya
Digital clothing twin (Riedelsheimer, Dorfhuber, and Stark 2020)	Consumers	Inform sustainable purchase decisions	+			Clothing	Proposition	Unspecified
		Provide enabling information for repair and care	+					
		Provide enabling information for resell and recycle	+					
Material passport (Honic et al. 2019)	Multiple stakeholders	Building materials management for reuse/recycling	+			Building	Proof-of-concept	Austria
Material passport (Honic, Kovacic, and Rechberger 2019)	Multiple stakeholders	Support design optimisation	+			Building	Proof-of-concept	Austria
Material passport (Smeets, Wang, and Drewniok 2019)	Stockists and fabricators	Reduce financial barriers for structural steel reuse	+		+	Building	Proposition	Europe
Building material passport (Munaro et al. 2019)	Unspecified	Building materials management for reuse/recycling	+			Building	Proof-of-concept	Brazil
Materials passport (Luscuere 2017)	Product designers	Drive innovation by providing product design guidance	+			Building	Proposition	Europe (the BAMB project)
	Property related businesses	Enable new business models that could benefit from reversibly designed products and systems	+		+			
	Multiple stakeholders	Enable value creation by providing information related to material health, deinstallation, disassembly, position, location, reverse logistics and so on	+		+			
Materials passport (Debacker, Manshoven, and Denis 2016)	Unspecified	Avoid down-cycling and enable high-quality recycling in construction and demolition waste management	+		+	Building	Proposition	Europe (the BAMB project)
	Building owners	Reduce waste and cost in renovation, refurbishment, and rebuilding	+		+			
	Building owners	Reduced cost in building maintenance management	+		+			
Materials passport (3 XN Adepa, 2016; as cited in Munaro and Tavares 2021)	Multiple stakeholders	Increase traceability of materials in terms of volume, location, and other specifications	+			Building	Proposition	Unspecified
	Multiple stakeholders	Enable operations such as disassembly, removal, repair, maintenance, and replacement services.	+					
Product passport (Portillo-Barco and Charnley 2015)	Manufacturers and product users	Manage hazardous and critical materials by tracking them through different reuses.	+			Aerospace product	Proof-of-concept	Unspecified
	Manufacturers and product users	Regenerate end-of-life products	+					
	Manufacturers and product users	Predictive performance and maintenance based on sensing data of environmental and operational conditions.			+			

(*Continued*)

Table A2. Continued.

DPP variant and reference	Stakeholder	Supply chain activity	Affected sustainability dimension			Product type	Nature of use case	Country
			ENV	SOC	ECO			
Circularity passport (EPEA, 2015; as cited in Munaro and Tavares 2021)	Manufacturers	Protect authentic producers against counterfeiting, tampering and misuse.		+	+	Unspecified	Proposition	Europe
	Secondary materials market	Improve control of material value streams; increase residual value; use secondary materials with known and defined content.	+		+			
Nutrient certificates (Hansen, Braungart, and Mulhall 2013)	Material producers, users, and reprocessors	Value chain enhancement by converting 'waste' into a more defined marketable resource.	+		+	Building	Proposition	Europe
	Manufacturers	Authentication and protection from industrial counterfeiting, adulteration, and diversion.		+	+			
	Manufacturers	Enhance supply security by providing a basis for reliable material recovery			+			
	Manufacturers and waste management companies	Foster value partnerships to maximise value recovery and financial benefits	+		+			
	Recyclers	Improve quality assurance and risk management standards by providing data on the conditions of recycled materials which enable intelligent materials pooling.	+		+			
	Multiple stakeholders	Improve tracking of resources, especially rare materials, between regions in a globalised economy	+					
Recycling passport (Spengler and Schröter 2003)	Agfa-Gevaert AG and its suppliers and recyclers	Provide product data to recyclers to develop efficient disassembly and recycling strategies	+		+	Electrical and electronic equipment	Real-world application	Germany

Note: ENV-Environmental; SOC-Social; ECO-Economical.